Major Assignment - CS771
MLR-56 Neural Nomads

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Question 1

We want a model of the form (for a single datapoint)

$$\hat{y} = p^\top \phi(z) \cdot x + b,$$

where $\phi(z)$ is the feature map for the polynomial kernel

$$K(z_1, z_2) = \phi(z_1)^\top \phi(z_2) = (z_1^\top z_2 + c)^d.$$

Goal: convert this into a pure kernel regression problem. That means we need to find a feature map $\psi(x, z)$ and kernel $\widetilde{K}$ such that there exists some $\tilde{p}$ satisfying

$$\tilde{p}^\top \psi(x, z) = p^\top \phi(z) \cdot x + b.$$

A simple and natural choice of ψ is to concatenate a scaled version of $\phi(z)$ and a constant feature for the bias. Concretely, define

$$\psi(x, z) = x\, \phi(z)1, \qquad \tilde{p} = pb.$$

Then

$$\tilde{p}^\top \psi(x, z) = p^\top (x\, \phi(z)) + b \cdot 1 = p^\top \phi(z) \cdot x + b,$$

which is exactly the desired form.

Therefore, the kernel (inner product of the ψ's) is

$$\widetilde{K}\big((x_1, z_1), (x_2, z_2)\big) = \psi(x_1, z_1)^\top \psi(x_2, z_2) = (x_1 \phi(z_1))^\top (x_2 \phi(z_2)) + 1 \cdot 1.$$

Using $\phi(z_1)^\top \phi(z_2) = K(z_1, z_2)$, we get the closed form

$$\widetilde{K}\big((x_1, z_1), (x_2, z_2)\big) = x_1 x_2\, K(z_1, z_2) + 1.$$

If the polynomial kernel is

$$K(z_1, z_2) = (z_1^\top z_2 + c)^d,$$

then

$$\widetilde{K}\big((x_1, z_1), (x_2, z_2)\big) = x_1 x_2\, (z_1^\top z_2 + c)^d + 1.$$

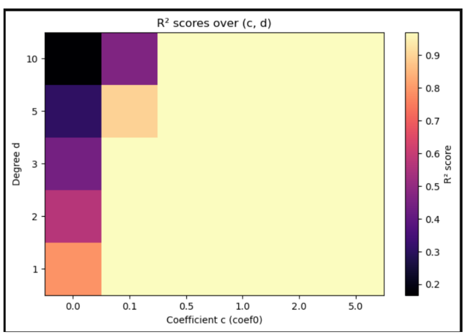
Question 2

We tested combinations of $c \in \{0.0, 0.1, 0.5, 1.0, 2.0, 5.0\}$ and $d \in \{1, 2, 3, 5, 10\}$ as hyperparameter values for the non-parametric regression model. The results are summarized below:

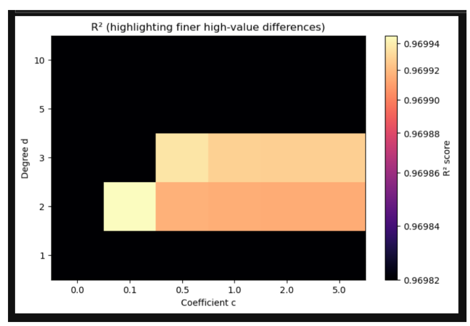
Table 1: R^2 scores for different (c, d) combinations.

Degree (d)	$c = 0.0$	$c = 0.1$	$c = 0.5$	$c = 1.0$	$c = 2.0$	$c = 5.0$
1	0.7898	0.9697	0.9697	0.9697	0.9697	0.9697
2	0.5714	**0.9699**	0.9699	0.9699	0.9699	0.9699
3	0.4404	0.9692	0.9699	0.9699	0.9699	0.9699
5	0.3073	0.8957	0.9698	0.9697	0.9697	0.9697
10	0.1669	0.4644	0.9689	0.9694	0.9694	0.9694

Both the normal and power-law normalized heatmaps illustrating the variation of the coefficient of determination (R^2) with respect to the kernel parameters c and d are given below.



Heatmap of R^2 scores.



Power-law normalized heatmap of R^2 scores.

Although the R^2 values for $c > 0.5$ are nearly identical to the maximum score, normalization reveals that the optimal performance occurs at a polynomial degree of $\boxed{d = 2}$ and a constant value of $\boxed{c = 0.1}$, where the model attains the highest R^2 score.

Question 3

Results

- **Train–train Gram matrix time:** 0.4024 seconds
- **Test–train Gram matrix time:** 1.6502 seconds
- **Kernel Ridge test R^2 score:** 0.9699

Question 4

We work with a XOR arbiter PUF built from two arbiter PUFs with $k = 32$ stages. For the *second* arbiter PUF (notation of the question: delays p_i, q_i, r_i, s_i), the standard intermediate parameters are

$$\alpha_i \triangleq \frac{p_i - q_i + r_i - s_i}{2}, \qquad \beta_i \triangleq \frac{p_i - q_i - r_i + s_i}{2}, \qquad 0 \le i \le k-1,$$

and the linear model $u \in \mathbb{R}^{k+1}$ is obtained as

$$u_0 = \alpha_0, \qquad u_i = \alpha_i + \beta_{i-1} \quad (1 \le i \le k-1), \qquad u_k = \beta_{k-1}.$$

An analogous construction (with delays a_i, b_i, c_i, d_i) gives a second model $v \in \mathbb{R}^{k+1}$ for the *first* arbiter PUF. The linear model of the XOR arbiter PUF is then

$$w \triangleq u \otimes v \in \mathbb{R}^{(k+1)^2},$$

where the Kronecker product is defined componentwise as

$$(u \otimes v)_{i(k+1)+j} = \sum_{a=0}^{k} \sum_{b=0}^{k} u_a v_b \, \delta_{ia} \, \delta_{jb}, \qquad 0 \le i, j \le k,$$

with δ_{ij} the Kronecker delta. Equivalently,

$$(u \otimes v) = (u_0 v_0, \ldots, u_0 v_k, \; u_1 v_0, \ldots, u_1 v_k, \; \ldots, \; u_k v_k).$$

Our task in Part 4 is: given $w \in \mathbb{R}^{(k+1)^2}$, construct non–negative delays

$$a_i, b_i, c_i, d_i, p_i, q_i, r_i, s_i \ge 0, \qquad 0 \le i \le k-1,$$

such that these delays generate exactly the same linear model w. Because the mapping from delays to w is non-injective, many solutions exist; we only need to describe one systematic construction.

Before decomposition, observe that for any nonzero scalar c,

$$(cu) \otimes ((1/c)v) = u \otimes v,$$

so the inverse is not unique. However, when w is reshaped into a $(k+1)\times(k+1)$ matrix,

$$W_{ij} \;\triangleq\; w_{\,i(k+1)+j} = \sum_{a,b} u_a v_b\, \delta_{ia}\, \delta_{jb} = u_i v_j,$$

the result is a rank–one matrix. Recovering u and v therefore reduces to finding any rank–one factorization of W, for which the singular value decomposition provides a canonical choice.

Step 1: Recovering u and v via de-Kronecker

Reshape w into the matrix $W \in \mathbb{R}^{(k+1)\times(k+1)}$ by

$$W_{ij} = w_{\,i(k+1)+j}.$$

By the discussion above,

$$W = uv^\top,$$

so W has rank one. Compute the SVD

$$W = U\Sigma V^\top, \qquad \Sigma = \mathrm{diag}(\sigma_1, 0, \ldots, 0),$$

and define

$$u_{est} = \sqrt{\sigma_1}\, U_{:,0}, \qquad v_{est} = \sqrt{\sigma_1}\, V_{:,0}.$$

Then $W = u_{est} v_{est}^\top$ and hence $w = u_{est} \otimes v_{est}$. Negate both vectors if necessary to ensure non-negative final entries.

Step 2: Recovering α_i, β_i from the single–PUF models

Each single–PUF model satisfies

$$u_0 = \alpha_0, \qquad u_i = \alpha_i + \beta_{i-1}\ (1 \le i \le k-1), \qquad u_k = \beta_{k-1}.$$

A simple, valid inverse mapping is

$$\alpha_i = u_i \quad (0 \le i \le k-1), \qquad \beta_i = 0\ (0 \le i \le k-2), \qquad \beta_{k-1} = u_k.$$

Apply the same formulas to v_{est} to obtain $\tilde{\alpha}_i, \tilde{\beta}_i$.

Step 3: Constructing non–negative delays

The stage delays satisfy

$$\alpha_i = \frac{p_i - q_i + r_i - s_i}{2}, \qquad \beta_i = \frac{p_i - q_i - r_i + s_i}{2}.$$

Introduce a slack variable $t_i \ge 0$ and set

$$p_i = t_i + \alpha_i + \beta_i, \quad q_i = t_i, \qquad r_i = t_i + \alpha_i - \beta_i, \quad s_i = t_i.$$

Substituting verifies that these delays recover α_i, β_i exactly. Choose

$$t_i = \max\{0,\ -(\alpha_i + \beta_i),\ -(\alpha_i - \beta_i)\}$$

to ensure non-negativity. Applying the same construction to $\tilde{\alpha}_i, \tilde{\beta}_i$ yields a_i, b_i, c_i, d_i.

Summary

Reshape w into W, extract u and v via the rank–one SVD factorization, convert these into α, β, and finally construct non-negative delays using a slack-variable method. The resulting delays produce exactly the original XOR–PUF model w.

Question 5

Results

- **my_decode runtime:** 0.00019 seconds
- **Euclidean distance to input linear model:** 4.34×10^{-16}